AI Glasses V2.3 Hardware Dimensions and Performance

Generated 2026-07-01 from current schematic/BOM and hardware_mechanical_database.yaml

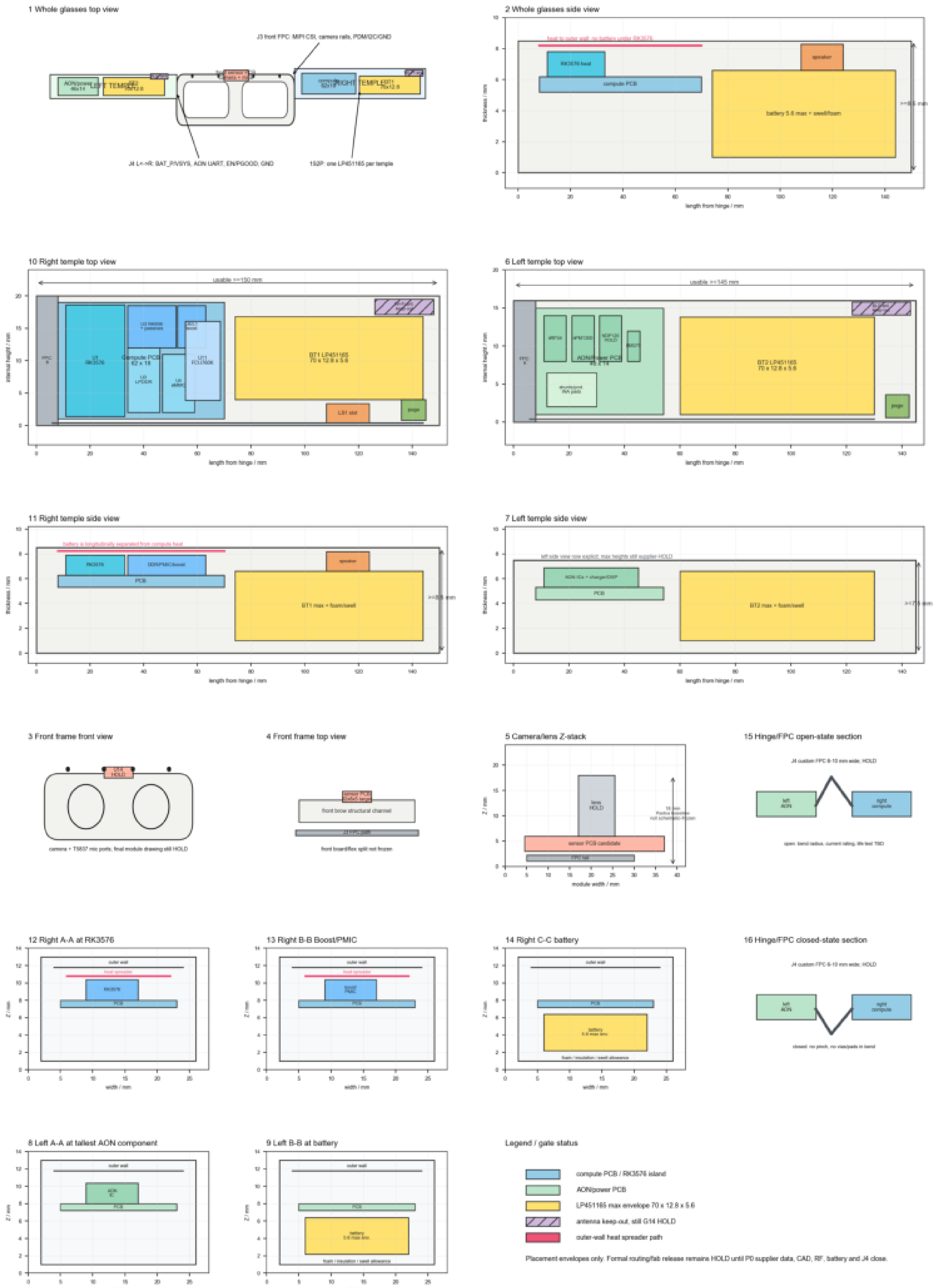
- Overall verdict before PCB layout: HOLD
- Thin optical frame: FAIL for current architecture
- Thick smart-glasses / sports frame: PASS WITH CONDITIONS after CAD/FPC/RF/battery proof
- AON and compute PCBs contain multiple ICs because those ICs are soldered to the board; this is allowed
- Real overlap of 3D envelopes, courtyards, battery swell, speaker cavity, or RF keep-out is not allowed
- Existing ERC report: 0 errors / 0 warnings; DRC not run because V2 has no PCB yet

Current power model

State	Model	Target/use
Deep Off / AON	22 mW	20-50 mW target
Quick Standby	134 mW	DDR/light sleep model
Phone assist	362 mW	3-5 h blended target
Mixed motion	450 mW	<=500-800 mW target
1080p record	1290 mW	45-90 min product target
AI + record	2065 mW	30-60 min / short burst target

16 required mechanical views - placement envelope atlas

AI Glasses V2.3 Mechanical Views - 16-view placement envelope atlas



Style updated from Phase 1.5 floorplan: mm axes, larger temple views, explicit side sections. Source: hardware_mechanical_database.yaml + current schematic/BOM. Not released PCB geometry.

Phase 1 design review answers

- Left temple side view is missing in the earlier render: HOLD until left side + A-A/B-B sections exist in CAD.
- Right temple side section is directionally correct but incomplete: add real max heights, solder, heat pad, insulation and tolerance stack.
- nRF54L15, nPM1300, NDP120 and BMI270 on one AON PCB is reasonable, with RF/thermal/vibration separation.
- AON 46 x 14 mm and Compute 62 x 18 mm are placement envelopes, not proof of complete circuitry.
- RK3576 + LPDDR routing needs 8-10 layer HDI review, ball maps, stack-up and DDR length report.
- Battery must stay longitudinally separated from RK3576/PMIC/boost; any real hot-source overlap is FAIL.
- Wi-Fi/BLE antenna keep-outs are HOLD until antenna SKU/tune and speaker/battery/screw clearance are proven.
- Cross-temple 1S2P and J4 current path are major HOLD items.

KiCad audit evidence generated

Audit item	Result	Evidence
Components	96	symbol_to_footprint_mapping.csv maps each schematic component to current footprint/status
Schematic nets	127	interconnect_matrix.csv/md records endpoints, domain crossings and layout rules
Footprint release	HOLD	VERIFY/HOLD/TBD/DNP items are NOT_FOR_FAB until official land patterns and package heights exist
ERC	Existing PASS	erc.report.txt shows 0 errors / 0 warnings on 2026-07-01T01:26:33; not rerun because kicad-cli unavailable
DRC / 3D	NOT RUN	v2 currently has no .kicad_pcb or verified STEP assembly

Decision

No KiCad schematic, footprint, PCB or 3D-model files were modified in this phase. That is intentional: the project rules require HOLD placeholders until exact MPNs, official package drawings, pinouts, land patterns and mechanical envelopes are available.

Component cards - simple language, source-bound dimensions

U1 RK3576 — HOLD

16.1 x 17.2 x TBD mm

main compute SoC

Voltage: RK806S rails from SOC_5V

Power: included in compute model / peak part of AI-burst compute row

Purpose: 运行 Linux/AI/编码/ISP 等高负载任务。

Placement: R-Temple Compute Board

Missing: G01 (RK3576 datasheet/HDG/ballmap/DDR guide)

U2 RK806S-5 — HOLD

TBD

SoC PMIC

Voltage: SOC_5V input; generates RK3576/DDR rails

Power: regulation loss included in compute model / peak TBD after rail-current budget

Purpose: 运行 Linux/AI/编码/ISP 等高负载任务。

Placement: R-Temple Compute Board

Missing: G03 (RK806S MPN/inductors/caps/timing per Radxa/Rockchip FAE)

U3 LPDDR4X 4 GB MPN TBD — HOLD

TBD

system memory

Voltage: RK806 DDR rails

Power: included in compute model / peak TBD by selected MPN

Purpose: 完成该功能模块在系统中的专用任务。

Placement: R-Temple Compute Board

Missing: G02 (LPDDR4X MPN/topology/placement, DDR review + length report)

U4 32 GB eMMC 5.1 MPN TBD — HOLD

TBD

boot/system storage

Voltage: VCC/VCCQ from RK806 rails

Power: included in compute model / peak TBD by selected MPN

Purpose: 完成该功能模块在系统中的专用任务。

Placement: R-Temple Compute Board

Missing: G04 (eMMC MPN + BSP/bootloader; cold-boot + power-loss recovery)

Component cards - simple language, source-bound dimensions

U11 FCU760KAAMD — HOLD

13.0 x 12.2 x 2.0 mm

Wi-Fi/Bluetooth module

Voltage: VBAT 3.0-3.6 V, typ 3.3 V

Power: radio average: 80-150 mW in current model / peak 353 mA @ 3.3 V max Tx current in short spec

Purpose: 完成该功能模块在系统中的专用任务。

Placement: R-Temple Compute Board

Missing: G05 (Quectel HW Design + RK3576 BSP driver/firmware/enum) + official LCC land pattern

U6 TPS61088 — HOLD

4.5 x 3.5 x TBD mm

main SoC boost

Voltage: VSYS in, SOC_5V out

Power: conversion loss in compute-island loss row / peak TBD from RK3576 boot/AI current

Purpose: 运行 Linux/AI/编码/ISP 等高负载任务。

Placement: R-Temple Compute Board

Missing: G06/G09 (measured RK3576 boot+AI peak, soft-start, droop, thermal)

L1 shielded low-DCR power inductor TBD — TBD

5.0 x 5.0 x 2.0 mm

main boost inductor

Voltage: SOC_5V boost switch node

Power: loss TBD by DCR/RMS current / peak Isat must cover boot/AI peak

Purpose: 完成该功能模块在系统中的专用任务。

Placement: R-Temple Compute Board

Missing: G06/G13 (Isat covers boot+AI peak; height <=2mm; thermal)

U12 TPS62825 — PROVISIONAL

1.5 x 1.5 x TBD mm

Wi-Fi buck regulator

Voltage: VSYS to WIFI_3V3

Power: ~5 mW loss allocation / peak sized for FCU760K Tx current

Purpose: 完成该功能模块在系统中的专用任务。

Placement: R-Temple Compute Board

Missing: G05/G06 (inductor/caps per module 353mA TX peak + ripple)

Component cards - simple language, source-bound dimensions

U20 MAX98360A — PROVISIONAL

TBD

digital class-D speaker amplifier

Voltage: AUDIO_PWR / I2S control

Power: audio average 2-3 mW electronics plus speaker output / peak depends on speaker impedance and acoustic target

Purpose: 输出提示音或语音。

Placement: R-Temple Compute Board

Missing: Verify Z/cavity/EMI/peak power

Y1 24 MHz 10 ppm crystal TBD — PROVISIONAL

3.2 x 2.5 x TBD mm

main clock

Voltage: oscillator pins on RK3576

Power: passive / peak passive

Purpose: 完成该功能模块在系统中的专用任务。

Placement: R-Temple Compute Board

Missing: G01 (reuse reference CL/ESR + placement distance per HDG)

U5 MX25U6432F — DNP

TBD

optional boot flash

Voltage: 1.8 V SPI/QSPI

Power: DNP / peak DNP

Purpose: 完成该功能模块在系统中的专用任务。

Placement: R-Temple Compute Board

Missing: Boot flow decides whether production needs it

BT1 LP451165 300 mAh — HOLD

70.0 x 12.8 x 5.6 mm

right battery cell

Voltage: 1S Li-Po nominal 3.7 V

Power: source / peak branch current TBD by 1S2P validation

Purpose: 提供整机电源和续航。

Placement: Temple Rears (batt/spkr/ant)

Missing: G07 (full datasheet, >=2C, IR, cycles, cert)

Component cards - simple language, source-bound dimensions

LS1 8 ohm 0.5-1 W micro speaker TBD — TBD

TBD

main speaker

Voltage: driven by MAX98360A

Power: 13-27 mW average in model / peak 0.5-1 W acoustic part class

Purpose: 输出提示音或语音。

Placement: Temple Rears (batt/spkr/ant)

Missing: MPN after acoustic cavity/volume/leak test

J1 magnetic pogo 4-6 pin TBD — TBD

10.0 x 3.0 x TBD mm

magnetic pogo charge/service

Voltage: USB_5V, GND, optional USB2/ID

Power: passive; contact I2R loss / peak charge/fault current rating TBD

Purpose: 连接板卡、电源、信号或充电/调试接口。

Placement: Temple Rears (batt/spkr/ant)

Missing: Magnet direction, sweat corrosion, short, cycle life

J3 FH26W series pin count TBD — HOLD

12.0 x 3.5 x 1.0 mm

front FPC connector

Voltage: MIPI/PDM/I2C/CAM rails/GND

Power: passive / peak pin-current TBD

Purpose: 连接板卡、电源、信号或充电/调试接口。

Placement: Front Sensor Board

Missing: G12 (pin count from final cam lane/mic/power split)

J4 custom 6-10 mm FPC / rigid-flex TBD — HOLD

TBD x 6-10 x 0.10-0.18 mm

left-right interconnect / future hinge FPC

Voltage: BAT_P/VSYS, GND, AON UART, EN/PGOOD/fault/status

Power: passive; I2R loss / peak must carry peak and fault current safely

Purpose: 连接板卡、电源、信号或充电/调试接口。

Placement: Temple Rears (batt/spkr/ant)

Missing: G12 (impedance, bend radius, life, hinge interference)

Component cards - simple language, source-bound dimensions

U7 nRF54L15 QFN48 candidate — PROVISIONAL

6.0 x 6.0 x 0.85 mm

always-on BLE MCU

Voltage: AON_1V8/AON_3V3

Power: 2-8 mW allocation depending state / peak radio peak TBD

Purpose: 在主 SoC 断电时保持 BLE、IMU、唤醒和电源控制。

Placement: L-Temple AON/Power Board

Missing: EVT-frozen (RF layout/SDK/package to freeze early)

U8 nPM1300 — HOLD

5.0 x 5.0 x TBD mm

AON PMIC/charger/fuel gauge

Voltage: BAT_P/USB_5V in; AON_1V8/AON_3V3 out

Power: 3-15 mW AON buck/quiescent allocation / peak charge/PMIC thermal TBD

Purpose: 在主 SoC 断电时保持 BLE、IMU、唤醒和电源控制。

Placement: L-Temple AON/Power Board

Missing: G08 (AON <=25/50mW) — configure via nPM PowerUP on EK before board

U9 NDP120 QFN40 candidate — HOLD

5.0 x 5.0 x TBD mm

always-on audio DSP

Voltage: AON rails, reset/wake to nRF54L15

Power: 12-20 mW project listening allocation / peak TBD by dev kit measurement

Purpose: 完成该功能模块在系统中的专用任务。

Placement: L-Temple AON/Power Board

Missing: Full datasheet + dev kit + measured listening power + NDA/licensing

U10 BMI270 — PROVISIONAL

2.5 x 3.0 x 0.8 mm

IMU

Voltage: AON_1V8/AON_3V3

Power: 1 mW low-power motion budget / peak TBD by IMU mode

Purpose: 完成该功能模块在系统中的专用任务。

Placement: L-Temple AON/Power Board

Missing: Evaluate false-trigger under real frame vibration

Component cards - simple language, source-bound dimensions

U23 BQ2970/BQ297xx suffix TBD — DNP

1.5 x 1.5 x 0.75 mm

optional 1S battery protection IC

Voltage: single-cell Li-ion protection

Power: 4 uA normal, 100 nA shutdown class / peak external FET path handles current

Purpose: 提供整机电源和续航。

Placement: L-Temple AON/Power Board

Missing: Keep ONE protection scheme; confirm suffix OVP/UVP/OCP vs nPM1300/boost UVLO (§17)

Q1 dual back-to-back N-MOSFET TBD — DNP

TBD

optional battery protection MOSFETs

Voltage: cell negative protection path

Power: DNP / peak RDS(on), VDS and pulse current TBD

Purpose: 提供整机电源和续航。

Placement: L-Temple AON/Power Board

Missing: Vds/RDSon/Vgs/peak per BQ2970 app (§17)

RS1 10 milliohm 1% 1 W shunt TBD — PROVISIONAL

TBD

total pack current shunt

Voltage: BAT_P/NPM_VBAT Kelvin sense

Power: I^2R loss / peak sized for pack current

Purpose: 完成该功能模块在系统中的专用任务。

Placement: L-Temple AON/Power Board

Missing: I_BAT_TOTAL; the one production-kept sense path

U25 INA238 — DNP

3.0 x 4.9 x TBD mm

optional total-current monitor

Voltage: AON I2C / shunt monitor

Power: 640 uA class when populated / peak 5 uA shutdown max class

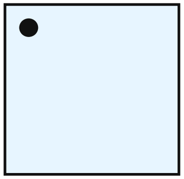
Purpose: 完成该功能模块在系统中的专用任务。

Placement: L-Temple AON/Power Board

Missing: Power Gate — EVT-A; production may keep this one

Component cards - simple language, source-bound dimensions

RT1 10k NTC curve TBD — PROVISIONAL



TBD

right-cell NTC

Voltage: AON analog / nPM1300 temperature input

Power: uW class divider / peak uW class

Purpose: 完成该功能模块在系统中的专用任务。

Placement: Temple Rears (batt/spkr/ant)

Missing: NTC curve matched to nPM1300 charger config

BT2 LP451165 300 mAh — HOLD

70.0 x 12.8 x 5.6 mm

left battery cell

Voltage: 1S Li-Po nominal 3.7 V

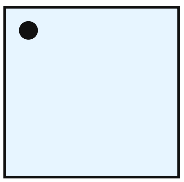
Power: source / peak branch current TBD by 1S2P validation

Purpose: 提供整机电源和续航。

Placement: Temple Rears (batt/spkr/ant)

Missing: G07

J6 2.4 GHz FPC/PCB antenna TBD — TBD



TBD

BLE antenna

Voltage: RF passive, 50 ohm

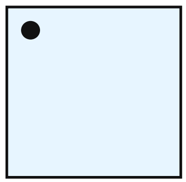
Power: passive / peak radio-dependent

Purpose: 把无线芯片的 RF 信号发射/接收出去。

Placement: Temple Rears (batt/spkr/ant)

Missing: Worn-state tuning; clear of battery/metal

U14 camera module — HOLD



TBD

camera module

Voltage: CAM_1V1, CAM_1V8_SW, CAM_2V9

Power: 250-300 mW camera row in current model / peak bare-sensor/module rail currents TBD

Purpose: 采集画面或给相机供电/保护。

Placement: Front Sensor Board

Missing: G10 (module lens/FOV/FPC pinout/lane/supply/timing from vendor)

Component cards - simple language, source-bound dimensions

CAM_LENS Radxa Camera 4K AS001 lens candidate — HOLD

32.5 x 32.5 x 18.0 mm

camera lens stack

Voltage: mechanical/optical

Power: passive / peak passive

Purpose: 采集画面或给相机供电/保护。

Placement: Virtual mechanical item

Missing: close gate/source data before PCB layout

MK1 T5837 — PROVISIONAL

3.5 x 2.65 x 0.98 mm

wake microphone

Voltage: AON microphone rail/PDM

Power: 3 mW wake-mic budget / peak TBD by mode

Purpose: 完成该功能模块在系统中的专用任务。

Placement: Front Sensor Board

Missing: G11 (mic coords/ports/wind/wake + AEC/beamforming)

MK2 T5837 — PROVISIONAL

3.5 x 2.65 x 0.98 mm

array microphone 1

Voltage: camera/compute audio PDM rail

Power: array mic share in active states / peak TBD by mode

Purpose: 完成该功能模块在系统中的专用任务。

Placement: Front Sensor Board

Missing: G11

MK3 T5837 — PROVISIONAL

3.5 x 2.65 x 0.98 mm

array microphone 2

Voltage: camera/compute audio PDM rail

Power: array mic share in active states / peak TBD by mode

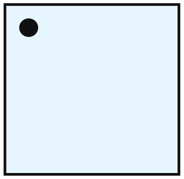
Purpose: 完成该功能模块在系统中的专用任务。

Placement: Front Sensor Board

Missing: G11

Component cards - simple language, source-bound dimensions

U15 TPS62840 — PROVISIONAL



TBD

camera 1.1 V buck

Voltage: CAM_1V1

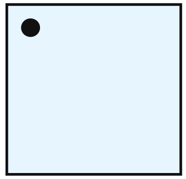
Power: 15-18 mW loss allocation when camera active / peak DVDD current plus margin

Purpose: 采集画面或给相机供电/保护。

Placement: Front Sensor Board

Missing: Output current/noise per final module (DVDD ~250mA max + margin)

U16 TLV75529PDRVR — PROVISIONAL



2.0 x 2.0 x TBD mm

camera 2.9 V LDO

Voltage: CAM_2V9

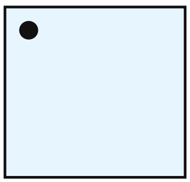
Power: 14-16 mW loss allocation when camera active / peak IMX415 AVDD current plus thermal

Purpose: 采集画面或给相机供电/保护。

Placement: Front Sensor Board

Missing: Verify vs IMX415 AVDD peak (156mA) + PSRR

U17 TPS22917DBVR — PROVISIONAL



2.9 x 2.8 x TBD mm

camera 1.8 V load switch

Voltage: AON_1V8 to CAM_1V8_SW

Power: ~1 mW loss allocation / peak IOVDD current, reverse-block/QOD check

Purpose: 采集画面或给相机供电/保护。

Placement: Front Sensor Board

Missing: Check reverse block, ramp, QOD, logic level

U18 TPD4E05U06 — PROVISIONAL

2.5 x 1.0 x TBD mm

MIPI/FPC ESD array 1

Voltage: MIPI/low-cap ESD

Power: leakage only / peak ESD clamp

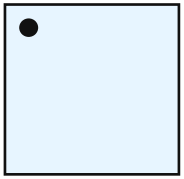
Purpose: 连接板卡、电源、信号或充电/调试接口。

Placement: Front Sensor Board

Missing: Low-cap array near FPC entry

Component cards - simple language, source-bound dimensions

CAM_FPC custom camera FPC TBD HOLD



TBD

camera FPC tail

Voltage: MIPI CSI, I2C, rails, GND

Power: passive; I2R/insertion loss / peak pin-current TBD

Purpose: 采集画面或给相机供电/保护。

Placement: Virtual mechanical item

Missing: close gate/source data before PCB layout

P0 open mechanical data before PCB layout

- RK806S-5 package / height: exact package/height missing; closure: Official package drawing + land pattern + rail sequence reviewed

- LPDDR4X exact MPN / size / ball map: memory MPN not selected; closure: MPN + package + ball map + DDR topology + length report

- eMMC exact MPN / package: eMMC MPN not selected; closure: MPN + package + BSP boot validation

- IMX415 lens / PCB / FPC / total Z: architecture mismatch; closure: Approved module drawing + lens/FPC/mounting + schematic/BOM update if changed

- Wi-Fi antenna: antenna SKU/keep-out not frozen; closure: Worn-state antenna tune + keep-out in CAD/PCB

- BLE antenna: antenna SKU/keep-out not frozen; closure: Antenna SKU + keep-out + matching + tune report

- main speaker: MPN/cavity/magnet unknown; closure: Speaker MPN + cavity + antenna magnet clearance passed

- magnetic pogo connector: target only; closure: MPN + pinout + current/fault/ESD validation

- hinge FPC: custom stack-up/pinout unknown; closure: Pinout + current rating + impedance + bend test

- Boost inductor: MPN/Isat/DCR/thermal unknown; closure: Selected MPN passes peak/droop/thermal

- FH26W exact pin count: final camera/audio/power split unknown; closure: Pin count/contact orientation + impedance checked

- battery pack drawing: supplier max/tolerance/swell not official; closure: Supplier drawing + dummy/real-cell fit test

- battery tab and cable exit: tab/cable direction unknown; closure: Pack drawing + strain-relief CAD

- battery NTC: B-value/package/placement unknown; closure: Curve + nPM1300 config + per-cell placement